AI & Software Engineering Workshop

Week 1, Day 5: Deploy and Ship

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From polling to webhook

- All week: **polling**, your laptop asks Telegram for updates
- Production: **webhook**, Telegram pushes to your server's URL
- PythonAnywhere runs the **same Flask app** as an always-on worker
- Same code, same `.env` idea: different delivery

Today your bot stops needing you.

The front door: `api/index.py`

Route	Job
`/api/health`	"Is the worker alive?" → `OK`
`/api/webhook`	Where Telegram delivers messages
`/api/deploy`	Pulls new code on push (we'll wire this up)

- The webhook checks a **secret token**: forged requests get 403. The bot generates and registers it itself on first boot.
- `threaded=False` in `bot/clients.py`: handlers must finish in the request, or the platform may kill them mid-reply.

Deploy: one command

1. Sign up at **pythonanywhere.com** (free Beginner tier)
2. Create an API token: *Account → API token*
3. Add to your local `.env`:

```
PA_USERNAME=<your PA username>
PA_API_TOKEN=<the token>
```

4. Ship it:

```
make deploy-pa
```

What just happened

The script created the web app, cloned **your fork**, built the venv, uploaded a production `.env`, including:

```
SQLITE_PATH=/home/<you>/bot.db
WEBHOOK_URL=https://<you>.pythonanywhere.com/api/webhook
```

`WEBHOOK_URL` = auto-registration: every boot, the bot tells Telegram "*deliver here.*" No manual `curl` needed.

Verify like an engineer

1. `https://<you>.pythonanywhere.com/api/health` → `OK`
2. Message your bot: the reply now comes from the server
3. **Close your laptop. Message it again.** 🎉
4. Something wrong? *Web tab → error log*: read the traceback

Push-to-deploy

One-time setup on your fork, in *Settings → Secrets → Actions*:

Secret	Value
`DEPLOY_SECRET`	random string (also in PA `.env`)
`PA_DEPLOY_URL`	`https://<you>.pythonanywhere.com/api/deploy`

Then:

```
git commit -am "New personality" && git push
```

~3 seconds later the live bot is updated. This is how real teams ship.

The fine print ⚠️

- PA free tier: click **"Run until 3 months from today"**-style renewal **every month**, or the app auto-disables
- PA emails you a week before. Set a phone reminder anyway.
- Free tier limits outbound internet to a whitelist: Telegram and Cerebras are on it

Final project: ship one original feature

- 🌍 Translator: Armenian ↔️ English on demand
- ❓ Quiz master: asks, scores, keeps streaks (you have a store!)
- 🎭 Mood reader: guesses the mood of a message
- 📝 Summarizer: paste a wall of text, get three lines

Or your own idea. It must be **live on your deployed bot** by demo time.

Demo day

1. Post your bot's username for the room
2. Try everyone's bot, find their feature
3. Vote: funniest persona, most useful feature, best surprise

Same bot as Tuesday morning, now it lives on the internet.

Next week

Your bot's brain is someone else's model behind an API.

Week 2: we train our own. 63 GB of Armenian text, one GPU, a 350-million-parameter model, built from scratch, fine-tuned to chat, and plugged into *this* bot.